

tivity challenge headlong and hopes to become one of the most connected states. In fact, the state wants to make Internet connectivity a basic human right.

Developed countries like Sweden and Canada made broadband connectivity a legal right many years ago. In 2016, the United Nations declared that depriving people of Internet connectivity was a human rights violation running contrary to international law. So, Kerala, which is known for being the most literate state, is working on the right lines. It plans to set up a new high-speed optical fibre network called K:Fon, which will run parallel to the existing electricity board network. Using this network, the government will provide free, subsidised or normal tariff Internet connectivity to all people based on their economic stature. Starting with at least one person from each family, Internet access will be gradually extended to all.

Telecom and start-ups at work. Indian telecom operators and start-ups are busy innovating technologies and business models for remote connectivity.

Vihaan Networks Limited (VNL), for example, believes in connecting remote communities using a base engine that is truly optimised and simple enough to be assembled and used by anyone. Their solar-powered WorldGSM Rural Site deploys a cascading star network architecture that enables rural wireless telephony and Internet access in areas with low average revenue per user (ARPU). With minimal capital and operating expenditure, this system gives mobile operators a decent return on investment, thereby enabling even rural operators to venture into this space.

AirJaldi service by class-A ISP Rural Broadband provides high-quality broadband connectivity to rural areas at reasonable rates. You might have read about them providing Wi-Fi services in India's first digital village Harisal. But, you will be in for a surprise if you look at the kind of remote regions they work in. It is difficult to even locate some of those regions in the Indian map.

RBB owns and operates networks in interior regions of states like

Himachal Pradesh, Uttarakhand, Jharkhand, Tamil Nadu, Maharashtra, Bihar and Karnataka. They manage to do this by looking at networks as 'retail ecologies.' On their website, they explain that mostly rural areas are subject to partial provisioning of Internet services, typified by the use of singular, non-scalable solutions (for example, VSAT, dongles, mobile devices connectivity) or customer-specific, costly and exclusive Wi-Fi connections (mostly banks and other enterprises capable of affording connectivity).

On the other hand, AirJaldi's philosophy and business model is based on creating networks that are meant, by design, to reach out to all potential clients at a reasonable price. Each relay is built to reach specific clients; each client is potentially a relay to other clients. They say that the ability to grow organically and quickly respond to demand increases the sustainability of their networks, and enables them to reach a large number of users in rural areas.

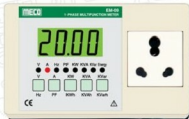
Arriving at the right mix. One thing is obvious. Governments need to look at a mix of technologies like fibre-optic cables, mobile networks, satellites and white space to achieve 100 per cent connectivity. As in the case of AirJaldi, the implementation needs to be flexible and capable of growing organically. There is no one-size-fits-all solution. The right mix depends on the distance from base stations, terrain, weather conditions, population density of the region, etc. Solutions must be designed such that these are economically viable and sustainable too.

So, as in any people-oriented effort, technology is only an enabler here too. The technology elements are there, and being developed and improved by the hour. But, the real challenge is to identify and string together the right technologies in a sustainable way. This is especially true for a country like ours, which has too much diversity in everything from terrain and economic status to weather and efficiency of local implementers. Let's wait and see how deep inside this complex country the Internet can penetrate! **EPY**



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